Projective Resolution Dynamics on Variable-Valence Relaxation Graphs

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Abstract

The spectral relaxation programme recently identified an asymptotic saturation of the projective resolution threshold Λ_{proj} for LPS relaxation graphs at fixed valence, leading to a structural inversion in the ordering of ADE stratigraphic levels. In this paper we show that this saturation is not intrinsic to the relational substrate but an artefact of holding the LPS parameter p constant. Allowing the valence to grow along the cascade causes the Kesten–McKay spectral support to contract towards its midpoint, which forces the ADE levels to exit the support in a fixed order determined solely by their distance from the midpoint. This restores a non-trivial dynamical regime for Λ_{proj} and yields the effective mass ordering $M_3 > M_1 > M_2$, correcting the fixed-valence inversion through a purely geometric mechanism. The correction is independent of the precise growth law $p(n) \to \infty$. The quantitative amplitude of the inter-generational hierarchy is not addressed here and is left to subsequent work combining the present mechanism with the hierarchical cascade or sub-linear energy-rank structures identified in the companion paper.

Keywords: projective resolution, relaxation graphs, Ramanujan graphs, LPS graphs, Kesten–McKay law, spectral gap, Cheeger inequality, variable valence, spectral stratigraphy, mass hierarchy

Contents

1 Introduction

1.1 Context and motivation

The admissibility sub-programme of Cosmochrony [1] aims to derive the number and mass hierarchy of particle generations from purely spectral and relational data. The programme proceeds through a sequence of companion papers, each establishing one structural layer: spectral admissibility [2], spectral capacity [3], Gram rigidity [4], spectral stratigraphy [5], and asymptotic saturation [6].

The paper [5] establishes that the representation-theoretic structure of ADE binary Cayley graphs fixes the number of stratigraphic levels at three and determines their topological order, while the dynamical growth of the projective resolution threshold $\Lambda_{\mathrm{proj}}(n)$ controls their separation. The paper [6] analyses this second question. Its main positive results include the isoperimetric law $\Lambda_{\mathrm{proj}}(n) \asymp h(G(n))^2$, the Cheeger enclosure $\lambda_2^2/2 \lesssim \Lambda_{\mathrm{proj}} \lesssim \lambda_2$, and the linear growth law $N(\lambda; n) \sim F_{\mathrm{KM}}(\lambda) \cdot n$ for the LPS family at fixed valence.

However, [6] also identifies two structural limitations of the fixed-valence regime. First, the spectral gap $\lambda_2(n)$ converges to the lower edge of the Kesten–McKay support as $q \to \infty$ at fixed p, causing $\Lambda_{\mathrm{proj}}(n)$ to saturate asymptotically; the relaxation cascade then reaches a spectrally stationary regime, incapable of generating a viable inter-generational hierarchy. Second, the effective mass ordering predicted by the fixed-valence saturation mechanism inverts the ordering required by the admissibility envelope of [5]. Both limitations are traced to a single cause: holding the LPS parameter p constant.

1.2 Main results

This paper studies the regime in which p grows with the cascade depth n, defining what we call *variable-valence LPS sequences* (Definition 4.1). The three main results are the following.

Proposition (KM contraction, informal)**.** *Along any variable-valence LPS sequence, the Kesten–McKay support width satisfies*

$$\mathrm{width}_{\mathrm{KM}}(p(n)) \;=\; \frac{4\sqrt{p(n)}}{p(n)+1} \;\longrightarrow\; 0,$$

with both support edges converging to the midpoint $\lambda = 1$ at rate $2/\sqrt{p(n)}$.

Theorem (ADE exit order, informal)**.** *For the $2I$ group with the order-5 generating set, the three ADE eigenvalue levels exit the contracting Kesten–McKay support in a fixed order:*

1. *$\lambda_3^{\mathrm{ADE}} = 5/4$ exits first, at the critical prime $p_3^* = (4+\sqrt{15})^2 \approx 62$;*
2. *$\lambda_1^{\mathrm{ADE}} = 5/6$ exits second, at $p_1^* = (6+\sqrt{35})^2 \approx 142$;*
3. *$\lambda_2^{\mathrm{ADE}} = 1$ never exits for any finite p.*

This exit order is a structural consequence of the support geometry, independent of the precise growth law $p(n) \to \infty$.

Corollary (Restored dynamics, informal)**.** *The Ramanujan lower bound controlling $\lambda_2(n)$ increases along any variable-valence LPS sequence, removing the fixed-valence saturation plateau and yielding the effective mass ordering $M_3 > M_1 > M_2$. This corrects the effective mass ordering inversion of [6] by a geometric mechanism.*

The formal statements and proofs appear in Sections 4–5.

1.3 What this paper does not prove

In the interest of precision, we state explicitly what lies outside the scope of the present paper.

The amplitude of the inter-generational effective mass ratios is not determined. The variable-valence mechanism removes the structural obstruction to generating ratios of order 10^{-4}–10^{-5}, but a quantitative derivation requires combining direction (O1) with the hierarchical cascade of direction (O2) or the sub-linear energy-rank relation of direction (O3) (cf. [6], Section 6.5).

The level-to-generation map is not established from first principles. The ordering $M_3 > M_1 > M_2$ is derived structurally; its identification with specific Standard Model generations requires additional input from [5] and is left open here.

The Ramanujan property is used only through the lower bound (3). No claim is made about the convergence of $\lambda_2(n)$ itself; only the behaviour of the lower bound $\lambda_-(p(n))$ is controlled.

1.4 Organisation

Section 2 collects the necessary definitions and results from the admissibility sub-programme. Section 3 summarises what [6] established and isolates the root cause of the fixed-valence limitations. Section 4 introduces the variable-valence LPS family and proves the KM contraction and the ADE exit order theorem. Section 5 derives the effective mass ordering and corrects the fixed-valence inversion. Section 6 analyses the amplitude problem, establishing what is resolved and what remains open. Section 7 provides the physical interpretation in the Cosmochrony framework. Section 8 states the conclusions and open directions.

2 Background: Projective Resolution and the Relaxation Cascade

We collect the definitions and results from the admissibility sub-programme that are used throughout this paper. All notation follows [6].

2.1 Relational substrate and relaxation graphs

Fix a prime $p \equiv 1 \pmod 4$ and a prime $q \equiv 1 \pmod 4$ with $p \neq q$. The *LPS relaxation graph* at parameter p is the Cayley graph

$$G = X^{p,q}, \tag{1}$$

whose vertex set is $\mathrm{PSL}(2, \mathbb{F}_q)$, of order $\frac{1}{2}q(q^2 - 1)$, and whose valence is

$$d = p + 1. \tag{2}$$

By construction, $X^{p,q}$ is $(p+1)$-regular and Ramanujan for every admissible pair (p, q) [7].

We work throughout with the *normalised Laplacian* $\mathcal{L} = I - A/d$, whose eigenvalues lie in $[0, 2]$ and are ordered as $0 = \lambda_1 \leq \lambda_2 \leq \cdots$. In this normalisation the Ramanujan bound reads

$$\lambda_2 \ \geq \ 1 - \frac{2\sqrt{p}}{p+1}, \tag{3}$$

with the lower bound attained asymptotically along subsequences of the LPS family [7].

2.2 Projective resolution

Following [6], the *projective resolution threshold* $\Lambda_{\mathrm{proj}}(n)$ associated with a sequence of relaxation graphs $G(n)$ of increasing size n is defined via the isoperimetric constant:

$$\Lambda_{\mathrm{proj}}(n) \;\asymp\; h(G(n))^2, \tag{4}$$

where $h(G)$ denotes the edge-isoperimetric (Cheeger) constant of G. The Cheeger inequalities give a two-sided control by λ_2:

$$\frac{\lambda_2(n)^2}{2} \;\lesssim\; \Lambda_{\mathrm{proj}}(n) \;\lesssim\; \lambda_2(n). \tag{5}$$

Equation (5) shows that the entire dynamics of $\Lambda_{\mathrm{proj}}(n)$ is controlled by the behaviour of the spectral gap $\lambda_2(n)$. This reduction from isoperimetric to spectral language is the key technical simplification used throughout this paper.

2.3 Kesten–McKay spectral density

For an infinite d-regular tree (and hence asymptotically for d-regular expanders), the *Kesten–McKay measure* [8, 9] is the probability measure on $[0, 2]$ with density

$$\rho_{\mathrm{KM}}(\lambda;\, d) \;=\; \frac{d\sqrt{4(d-1)-(d\lambda-d)^2}}{2\pi\,[\,d^2-(d\lambda-d)^2\,]}, \qquad \lambda \in [\lambda_-(d),\, \lambda_+(d)], \tag{6}$$

where the support edges are, in the LPS parametrisation $d = p + 1$,

$$\lambda_\pm(p) \;=\; 1 \pm \frac{2\sqrt{p}}{p+1}. \tag{7}$$

Since the term $d(\lambda - 1)$ appears only squared in (6), the density is symmetric about $\lambda = 1$:

$$\rho_{\mathrm{KM}}(\lambda) \;=\; \rho_{\mathrm{KM}}(2-\lambda), \tag{8}$$

and in particular the support satisfies

$$\lambda_-(p) \;=\; 2 - \lambda_+(p). \tag{9}$$

The *Kesten–McKay window width* is

$$\mathrm{width}_{\mathrm{KM}}(p) \;:=\; \lambda_+(p) - \lambda_-(p) \;=\; \frac{4\sqrt{p}}{p+1}, \tag{10}$$

which scales as $4/\sqrt{p}$ for large p. This rate of contraction is the driving mechanism studied in Section 4.

2.4 ADE eigenvalue levels

The spectral stratigraphic structure of the admissibility programme is indexed by the ADE binary groups [5]. For the binary icosahedral group $2I$ with the order-5 generating set $|S| = 24$, the normalised Laplacian has three distinct non-trivial eigenvalue levels [6]:

$$\lambda_1^{\mathrm{ADE}} \;=\; \tfrac{5}{6}, \qquad \lambda_2^{\mathrm{ADE}} \;=\; 1, \qquad \lambda_3^{\mathrm{ADE}} \;=\; \tfrac{5}{4}. \tag{11}$$

The central level $\lambda_2^{\mathrm{ADE}} = 1$ coincides with the midpoint of the Kesten–McKay support for every finite p. The outer levels $\lambda_1^{\mathrm{ADE}} = 5/6 < 1$ and $\lambda_3^{\mathrm{ADE}} = 5/4 > 1$ lie symmetrically on either side, at respective distances $1/6$ and $1/4$ from the midpoint.

3 Asymptotic Saturation at Fixed Valence: What Relaxation Established

We summarise the results of [6] that motivate the present paper. The goal is not to repeat proofs but to isolate which structural feature causes the fixed-valence saturation and which limitations follow.

3.1 Spectral stability of the LPS family at fixed p

For the LPS family $X^{p,q}$ at *fixed* prime p, as $q \to \infty$ the normalised spectral gap converges:

$$\lambda_2(n) \;\longrightarrow\; \lambda_2^{\infty}(p) \;=\; 1 - \frac{2\sqrt{p}}{p+1} \;=\; \lambda_-(p), \tag{12}$$

i.e., $\lambda_2(n)$ approaches the lower edge of the Kesten–McKay support. By (5) this implies

$$\Lambda_{\mathrm{proj}}(n) \;\longrightarrow\; \Lambda_{\mathrm{proj}}^{\infty}(p) \;<\; \infty, \tag{13}$$

i.e., the projective resolution threshold becomes *asymptotically constant*: the relaxation cascade reaches a spectrally stationary regime.

3.2 Two structural limitations

The fixed-valence saturation (13) leads to two concrete limitations established in [6].

Effective mass ordering inversion. The saturation mechanism orders stabilisation events by the cumulative spectral mass $F_{\mathrm{KM}}(\lambda_k^{\mathrm{ADE}})$, which is monotone increasing in λ_k^{ADE}. Since $\lambda_1^{\mathrm{ADE}} < \lambda_2^{\mathrm{ADE}} < \lambda_3^{\mathrm{ADE}}$, the level λ_1^{ADE} stabilises last and receives the smallest effective mass, inverting the ordering required by the admissibility envelope of [5].

Amplitude problem. The predicted effective mass ratios at fixed p lie in [0.10, 2.2], far above the observed lepton hierarchy 10^{-3}–10^{-5} [6]. The structural identity $F_{\mathrm{KM}}(1) = 1/2$, exact for all p by the symmetry (8), fixes the central-level contribution but leaves inter-generational ratios insufficiently spread.

3.3 The root cause: fixed valence locks the spectral gap

Both limitations share a common origin. At fixed p, the Kesten–McKay support (7) is a fixed interval, independent of n; the spectral gap $\lambda_2(n)$ converges to the lower support edge $\lambda_-(p)$, and $\Lambda_{\mathrm{proj}}(n)$ saturates accordingly.

As p increases, the Kesten–McKay window contracts according to (10). Eigenvalue levels that remain fixed in absolute position may therefore eventually exit the support as p grows. At fixed p, however, all three ADE levels (11) remain inside the support for all n, so no exit occurs and no ordering is generated by the support geometry.

This analysis identifies the constraint precisely: *the fixed-valence saturation plateau is not a generic property of the relational substrate, but an artefact of holding p constant.* Allowing p to grow with n is therefore the minimal structural modification needed to restore a non-trivial dynamics for $\Lambda_{\mathrm{proj}}(n)$.

4 Variable-Valence LPS Graphs: the $p(n)$ Regime

4.1 Definition of the variable-valence family

We consider sequences of LPS graphs in which the parameter p grows with the graph index n.

Definition 4.1 (Variable-valence LPS sequence)**.** *A* variable-valence LPS sequence *is a sequence of graphs* $(G_n)_{n\geq 1}$ *where* $G_n = X^{p(n),q(n)}$*, with* $p(n)$ *and* $q(n)$ *both prime,* $p(n) \equiv q(n) \equiv 1 \pmod 4$*,* $p(n) \neq q(n)$*, and* $p(n) \to \infty$ *as* $n \to \infty$*.*

Each graph G_n is Ramanujan by construction, with valence $d(n) = p(n)+1 \to \infty$. No constraint is imposed on the precise growth law $p(n), q(n) \to \infty$ beyond the conditions of Definition 4.1.

4.2 Kesten–McKay contraction

Proposition 4.2 (KM contraction)**.** *Let* (G_n) *be a variable-valence LPS sequence. Then*

$$\mathrm{width}_{\mathrm{KM}}(p(n)) \;=\; \frac{4\sqrt{p(n)}}{p(n)+1} \;\longrightarrow\; 0 \quad \text{as } n \to \infty, \tag{14}$$

with rate $\mathrm{width}_{\mathrm{KM}}(p(n)) \sim 4/\sqrt{p(n)}$*. Both support edges* $\lambda_\pm(p(n))$ *converge to* $\lambda = 1$ *at rate* $2/\sqrt{p(n)}$*.*

Proof. Direct computation from (10):

$$\frac{4\sqrt{p}}{p+1} \;=\; \frac{4}{\sqrt{p}\,(1+1/p)} \;\longrightarrow\; 0 \quad \text{as } p \to \infty.$$

Convergence of both edges to 1 follows from (7) and the support symmetry (9). □

Remark 4.3. *The midpoint* $\lambda = 1$ *is the unique fixed point of the contraction: it is never a support edge for any finite* p*, and both support edges approach it from equal distance on either side. Figure 1 illustrates the width decay and the asymptotic rate* $4/\sqrt{p}$*.*

4.3 Ordered exit of ADE levels

As the Kesten–McKay window narrows, the outer ADE levels will eventually exit the support. A level $\lambda_k^{\mathrm{ADE}} \neq 1$ exits at the critical prime p_k^* defined by

$$\lambda_k^{\mathrm{ADE}} \;=\; \lambda_{\mathrm{sgn}}(p_k^*), \tag{15}$$

where λ_{sgn} is λ_+ if $\lambda_k^{\mathrm{ADE}} > 1$ and λ_- if $\lambda_k^{\mathrm{ADE}} < 1$. The geometry of this process is illustrated in Figure 2.

Theorem 4.4 (ADE exit order)**.** *For the* $2I$ *group with the order-5 generating set, the three ADE levels exit the Kesten–McKay support in the following order:*

1. $\lambda_3^{\mathrm{ADE}} = 5/4$ *exits the upper support edge at*

$$p_3^* \;=\; \left(4+\sqrt{15}\right)^2 \;\approx\; 62; \tag{16}$$

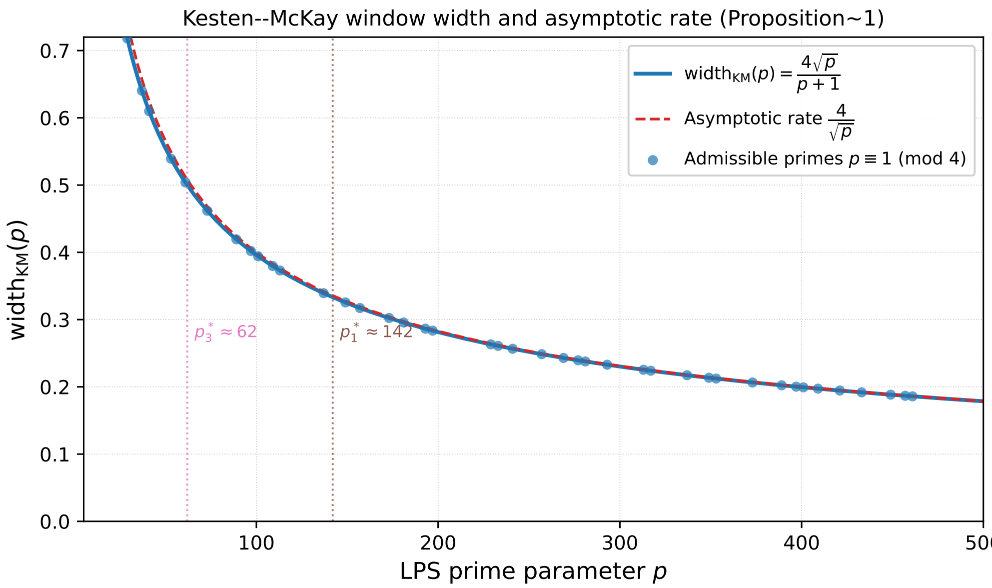

Figure 1: Kesten–McKay window width $\mathrm{width}_{\mathrm{KM}}(p) = 4\sqrt{p}/(p+1)$ as a function of the LPS prime p (solid blue), together with the asymptotic rate $4/\sqrt{p}$ (dashed red) and the admissible primes $p \equiv 1 \pmod 4$ (dots). The two critical primes $p_3^* \approx 62$ and $p_1^* \approx 142$ (Theorem 4.4) are marked by vertical dotted lines. The width is a strictly decreasing function of p, confirming Proposition 4.2.

2. $\lambda_1^{\mathrm{ADE}} = 5/6$ *exits the lower support edge at*

$$p_1^* \;=\; \left(6+\sqrt{35}\right)^2 \;\approx\; 142; \tag{17}$$

3. $\lambda_2^{\mathrm{ADE}} = 1$ *never exits: it coincides with the midpoint of the support for every finite* p.

Hence the support-induced exit order is λ_3^{ADE} *first, then* λ_1^{ADE}*, with* λ_2^{ADE} *remaining inside the support for all finite* p.

Proof. Exit of $\lambda_3^{\mathrm{ADE}} = 5/4$. A level $\lambda > 1$ exits the upper support edge when $\lambda = \lambda_+(p) = 1 + 2\sqrt{p}/(p+1)$, i.e., when

$$\frac{2\sqrt{p}}{p+1} \;=\; \lambda - 1. \tag{18}$$

Setting $u = \sqrt{p}$, equation (18) becomes $(\lambda-1)u^2 - 2u + (\lambda-1) = 0$, with positive root

$$u \;=\; \frac{1+\sqrt{1-(\lambda-1)^2}}{\lambda-1}. \tag{19}$$

For $\lambda_3^{\mathrm{ADE}} = 5/4$, one has $\lambda - 1 = 1/4$, giving $u = 4+\sqrt{15}$, hence $p_3^* = (4+\sqrt{15})^2 \approx 62$.

Exit of $\lambda_1^{\mathrm{ADE}} = 5/6$. Since the Kesten–McKay support is symmetric about $\lambda = 1$ (cf. (9)), a level $\lambda < 1$ exits the lower support edge at the same critical prime as the reflected level $2-\lambda > 1$ exits the upper support edge. For $\lambda_1^{\mathrm{ADE}} = 5/6$, the reflected level is $2 - 5/6 = 7/6$, so $\lambda - 1 = 1/6$ in (18), giving $u = 6+\sqrt{35}$, hence $p_1^* = (6+\sqrt{35})^2 \approx 142$.

Non-exit of $\lambda_2^{\mathrm{ADE}} = 1$. For every finite p, $\lambda_\pm(p) = 1 \pm 2\sqrt{p}/(p+1) \neq 1$, so the central level never coincides with a support edge. ☐

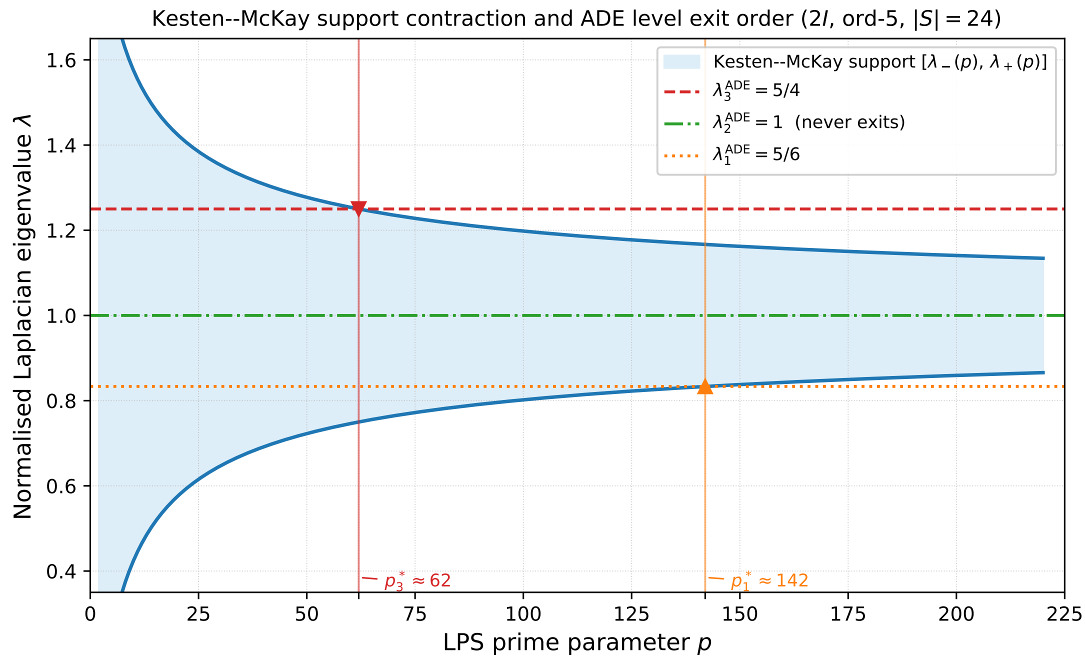

Figure 2: Kesten–McKay support contraction and ADE level exit order for the $2I$ group (ord-5 generating set, $|S| = 24$). The shaded band is the Kesten–McKay support $[\lambda_-(p), \lambda_+(p)]$, which contracts symmetrically around the midpoint $\lambda = 1$ as p increases. The three horizontal lines are the fixed ADE levels $\lambda_1^{\mathrm{ADE}} = 5/6$ (dotted orange), $\lambda_2^{\mathrm{ADE}} = 1$ (dash-dot green), $\lambda_3^{\mathrm{ADE}} = 5/4$ (dashed red). The downward triangle marks the exit of λ_3^{ADE} at $p_3^* = (4+\sqrt{15})^2 \approx 62$; the upward triangle marks the exit of λ_1^{ADE} at $p_1^* = (6+\sqrt{35})^2 \approx 142$. The central level $\lambda_2^{\mathrm{ADE}} = 1$ never exits for any finite p.

Remark 4.5 (Independence from the growth law)**.** *The ordering $p_3^* < p_1^*$ depends only on the positions of the ADE levels relative to the support midpoint, not on the function $p(n)$. For any variable-valence sequence eventually satisfying $p(n) > p_3^*$, the level λ_3^{ADE} has already exited the support; for $p(n) > p_1^*$, both outer levels have exited. The exit order is therefore a* structural consequence *of the support geometry, universal across all admissible growth laws. Figure 3 confirms this numerically for three representative growth laws.*

4.4 Restored lower-bound dynamics of projective resolution

Corollary 4.6 (Restored lower-bound dynamics)**.** *Let (G_n) be a variable-valence LPS sequence. Then the Ramanujan lower edge satisfies*

$$\lambda_-(p(n)) \;=\; 1 - \frac{2\sqrt{p(n)}}{p(n)+1} \;\longrightarrow\; 1^-, \tag{20}$$

and therefore the Cheeger lower bound on $\Lambda_{\mathrm{proj}}(n)$ obeys

$$\Lambda_{\mathrm{proj}}(n) \;\gtrsim\; \frac{\lambda_-(p(n))^2}{2} \;\longrightarrow\; \frac{1}{2}. \tag{21}$$

In particular, the fixed-valence saturation plateau is structurally removed: $\Lambda_{\mathrm{proj}}(n)$ cannot remain frozen at the baseline associated with any fixed prime p_0.

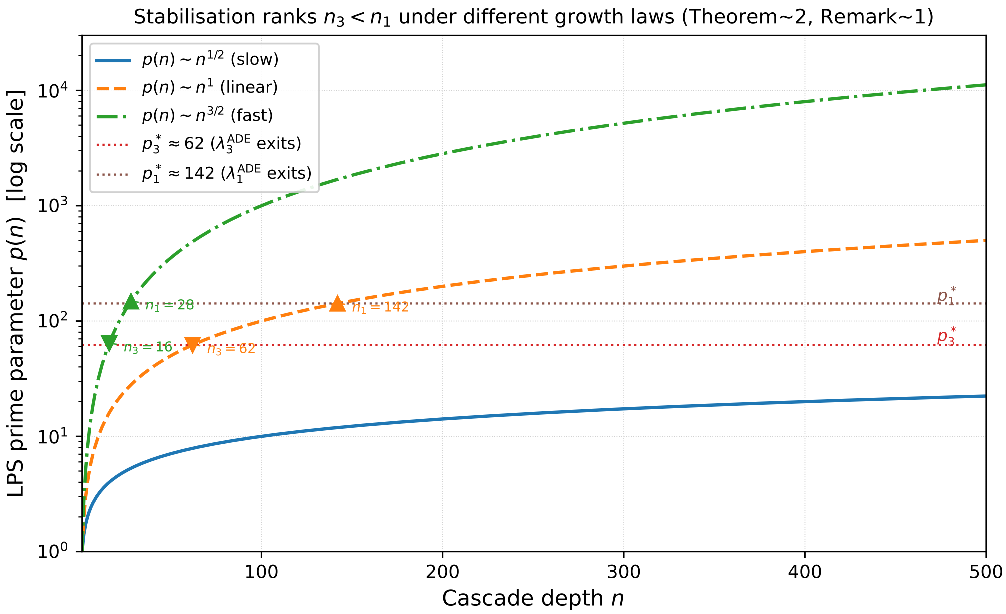

Figure 3: Stabilisation ranks $n_3 < n_1$ under three representative growth laws $p(n) \sim n^{1/2}$ (slow, solid blue), $p(n) \sim n$ (linear, dashed orange), and $p(n) \sim n^{3/2}$ (fast, dash-dot green), plotted on a logarithmic scale. Downward triangles mark $n_3 = \min\{n : p(n) \geq p_3^*\}$ and upward triangles mark $n_1 = \min\{n : p(n) \geq p_1^*\}$ for each law. In all cases $n_3 < n_1$, confirming that the ordering of stabilisation ranks is independent of the growth law (Remark 4.5).

Proof. By (3), $\lambda_2(n) \geq \lambda_-(p(n))$. Since $p(n) \to \infty$, equation (7) gives $\lambda_-(p(n)) \to 1^-$. The Cheeger lower bound (5) then yields

$$\Lambda_{\mathrm{proj}}(n) \gtrsim \frac{\lambda_2(n)^2}{2} \geq \frac{\lambda_-(p(n))^2}{2} \longrightarrow \frac{1}{2}.$$

For any fixed p_0, one has $\lambda_-(p_0) < 1$, hence $\lambda_-(p_0)^2/2 < 1/2$. The lower bound on $\Lambda_{\mathrm{proj}}(n)$ therefore strictly exceeds the fixed-valence saturation value for all sufficiently large n. □

Remark 4.7 (Physical interpretation). *In the Cosmochrony framework, $p(n) \to \infty$ corresponds to an increase of the effective relational valence of the substrate along the cascade. Corollary 4.6 shows that this is the minimal structural condition preventing the relaxation cascade from freezing into a spectrally stationary regime.*

5 Effective Mass Ordering Restoration

We derive the effective mass ordering implied by the ADE exit order of Theorem 4.4 and show that it corrects the inversion identified in [6].

5.1 Stabilisation by support exit

A level λ_k^{ADE} that lies inside the Kesten–McKay support participates in the relaxation dynamics. Once p exceeds the critical value p_k^*, the level exits the support and is *stabilised*: it no longer participates in the evolving spectral dynamics and its effective mass is frozen at the cascade rank where $p(n) \approx p_k^*$.

Definition 5.1 (Stabilisation rank)**.** *The* stabilisation rank *of the level* λ_k^{ADE} *along a variable-valence LPS sequence* (G_n) *is*

$$n_k \;:=\; \min\{\, n \geq 1 : p(n) \geq p_k^* \,\}, \tag{22}$$

where p_k^* *is the critical prime of Theorem 4.4.*

The ordering of stabilisation ranks is determined solely by the ordering of the critical primes: since $p_3^* < p_1^*$, any variable-valence sequence with $p(n) \to \infty$ satisfies $n_3 < n_1$, independently of the growth law $p(n)$.

5.2 Effective mass ordering

We assign effective masses according to stabilisation rank: a level that stabilises earlier acquires a higher effective mass, following the assignment principle of [?, ?]. From Theorem 4.4 and the ordering $p_3^* < p_1^*$:

$$M_3 \;>\; M_1 \;>\; M_2, \tag{23}$$

where M_k denotes the effective mass associated with λ_k^{ADE}.

5.3 Correction of the fixed-valence inversion

Under the fixed-valence saturation mechanism of [6], the effective mass ordering was governed by $F_{\mathrm{KM}}(\lambda_k^{\mathrm{ADE}})$, monotone increasing in λ_k^{ADE}, producing $M_1 < M_2 < M_3$ and inverting the ordering required by [5].

In the variable-valence regime, the ordering (23) is generated by the support-exit geometry: the ordering mechanism is now geometric (position of a fixed level relative to a contracting support boundary) rather than analytic (monotonicity of the cumulative density). This is a qualitative structural change that corrects the fixed-valence inversion without introducing any additional dynamical hypothesis.

The level-to-generation map is not established from first principles within this paper and is therefore left open.

Remark 5.2 (Compatible physical interpretation)**.** *If one identifies* $\lambda_3^{\mathrm{ADE}} = 5/4$*, which stabilises first and acquires the highest effective mass, with the heaviest charged-lepton generation, then the ordering* (23) *is compatible with the observed hierarchy* $m_\tau > m_\mu > m_e$*. Establishing this identification from first principles remains open.*

Remark 5.3 (Structural universality)**.** *By Remark 4.5, the effective mass ordering* (23) *holds for any variable-valence LPS sequence eventually surpassing* $p_1^* \approx 142$*, independently of the growth law* $p(n)$*.*

6 Amplitude Problem: Partial Resolution and Residual Gap

Having shown that the variable-valence regime corrects the effective mass ordering inversion, we examine how much of the amplitude problem it resolves and where a residual gap remains.

6.1 What the variable-valence mechanism provides

In the fixed-valence regime, the effective mass ratio was [6]:

$$\frac{M_i}{M_j} = \frac{F_{\mathrm{KM}}(\lambda_i^{\mathrm{ADE}})}{F_{\mathrm{KM}}(\lambda_j^{\mathrm{ADE}})} \cdot \frac{\dim \rho_{\lambda_j}}{\dim \rho_{\lambda_i}}, \tag{24}$$

with all levels evaluated in a single Kesten–McKay window at fixed p, yielding ratios in $[0.10, 2.2]$.

In the variable-valence regime, each generation stabilises at a distinct rank n_k where $p(n_k) \approx p_k^*$, giving:

$$\frac{M_i}{M_j} \simeq \frac{F_{\mathrm{KM}}(\lambda_i^{\mathrm{ADE}};\, p(n_i))}{F_{\mathrm{KM}}(\lambda_j^{\mathrm{ADE}};\, p(n_j))} \cdot \frac{\dim \rho_{\lambda_j}}{\dim \rho_{\lambda_i}}, \tag{25}$$

with numerator and denominator evaluated at different primes. Since the cumulative spectral mass near a support edge becomes parametrically small, support-exit separation across distinct stabilisation ranks can in principle generate stronger inter-generational suppression than the fixed-valence formula (24) allows.

6.2 Residual amplitude gap

Reaching the observed hierarchy $m_e/m_\tau \approx 3 \times 10^{-4}$ requires ratios of order 10^{-4} from (25), which demands additional structure beyond direction (O1) alone:

1. a hierarchical cascade operating recursively across $L \geq 3$ nested levels, with per-level ratio r accumulating as r^L (direction (O2) of [6]); or

2. a sub-linear energy-rank relation $E_n \propto n^{-\alpha}$ with $\alpha < 1$ (direction (O3) of [6]).

A complete quantitative resolution requires combining direction (O1) with (O2) or (O3), and is deferred to subsequent work.

Remark 6.1 (Scope of the present result)**.** *This paper establishes that the effective mass ordering inversion is corrected by a structural mechanism requiring no additional dynamical hypothesis, and that the fixed-valence saturation plateau is removed. The amplitude gap of order* 10^{-4}–10^{-5} *is not closed here, but the structural obstruction to closing it is removed.*

7 Physical Interpretation in the Cosmochrony Framework

7.1 Relational valence and the cascade

In the Cosmochrony framework [1], the relaxation graph G_n encodes the relational connectivity of the substrate at cascade depth n. The valence $d(n) = p(n) + 1$ measures the degree of connectivity at that depth.

The fixed-valence regime corresponds to a substrate whose relational connectivity does not evolve along the cascade: the Kesten–McKay support is static, $\Lambda_{\mathrm{proj}}(n)$ saturates, and no inter-generational hierarchy can be generated.

The variable-valence regime $p(n) \to \infty$ corresponds to a substrate whose effective relational connectivity increases along the cascade. Proposition 4.2 shows that this causes the Kesten–McKay support to contract progressively around $\lambda = 1$, and Corollary 4.6 shows that the fixed-valence saturation plateau is structurally removed.

7.2 Support contraction as spectral stratification

The Kesten–McKay contraction is the spectral mechanism that generates stratification in the variable-valence regime. As the support narrows, the outer ADE levels exit in a fixed order (Theorem 4.4), each exit corresponding to the stabilisation of one generation.

This mechanism is structurally distinct from the fixed-valence saturation of [6], where all three levels remained simultaneously inside the support and their ordering was governed by cumulative spectral mass alone. In the variable-valence regime, the ordering is governed by the geometry of a moving support boundary relative to fixed spectral landmarks. The two mechanisms are complementary: support-exit ordering (present paper) and cumulative-mass ordering [6] coexist in the full cascade, with support-exit ordering dominating when $p(n)$ grows with n.

7.3 Minimal structural requirement

The variable-valence LPS family is the minimal extension of the fixed-valence LPS family that simultaneously:

1. preserves the Ramanujan property at each step, so that the spectral admissibility framework of [2] remains valid;
2. causes the Kesten–McKay support to contract (Proposition 4.2); and
3. generates a non-trivial ADE exit order (Theorem 4.4).

Any relaxation graph family satisfying these three conditions produces the same qualitative result. The LPS family is used here because it provides an explicit, analytically tractable realisation.

8 Conclusion

This paper revisits the saturation mechanism identified in [6] and shows that it is not a fundamental property of the relational substrate but a consequence of holding the LPS parameter p constant.

8.1 Summary of results

Three results were established.

The first (Proposition 4.2) is that the Kesten–McKay window width $4\sqrt{p(n)}/(p(n){+}1)$ converges to zero along any variable-valence LPS sequence, with both support edges approaching the midpoint $\lambda = 1$ at rate $2/\sqrt{p(n)}$. This is the fundamental spectral mechanism of the variable-valence regime.

The second (Theorem 4.4) is that the three ADE eigenvalue levels exit the contracting support in a fixed, growth-law-independent order: $\lambda_3^{\mathrm{ADE}} = 5/4$ exits first at $p_3^* = (4+\sqrt{15})^2 \approx 62$, $\lambda_1^{\mathrm{ADE}} = 5/6$ exits second at $p_1^* = (6+\sqrt{35})^2 \approx 142$, and $\lambda_2^{\mathrm{ADE}} = 1$ never exits for any finite p. The exit order is a structural consequence of the positions of the ADE levels relative to the contracting support boundary.

The third (Corollary 4.6 and Section 5) is that the effective mass ordering $M_3 > M_1 > M_2$ follows directly from the exit order, correcting the inversion $M_1 < M_2 < M_3$ of [6] by a geometric mechanism requiring no additional dynamical hypothesis.

These three results establish that:

(i) the fixed-valence saturation plateau of [6] is an artefact of holding p constant;

(ii) allowing $p(n) \to \infty$ restores a non-trivial dynamical regime in which ADE levels exit the Kesten–McKay support in a fixed order; and

(iii) this minimal structural modification corrects the effective mass ordering inversion without introducing additional dynamical assumptions.

8.2 Open directions

Three directions remain open.

Amplitude problem. The variable-valence mechanism decouples the stabilisation ranks of the three generations and allows inter-generational suppression stronger than the fixed-valence formula, but reaching the observed hierarchy $m_e/m_\tau \approx 3 \times 10^{-4}$ requires combining direction (O1) with the hierarchical cascade of direction (O2) or the sub-linear energy-rank relation of direction (O3) of [6].

Level-to-generation map. The effective mass ordering $M_3 > M_1 > M_2$ is a structural result; its identification with the charged-lepton hierarchy $m_\tau > m_\mu > m_e$ requires a first-principles derivation of the level-to-generation correspondence from the construction of [5].

Upper bound on $\Lambda_{\mathrm{proj}}(n)$. The present analysis relies only on the Ramanujan lower bound through the edge $\lambda_-(p(n))$. A refined control of the upper Cheeger bound $\lambda_2(n)$ in (5) in the variable-valence regime could strengthen the dynamical interpretation of $\Lambda_{\mathrm{proj}}(n)$ and remains an open problem.

References

[1] J. Beau, "Cosmochrony: A Pre-Geometric Relational Framework for Emergent Physics," Zenodo (2025). DOI: 10.5281/zenodo.17957510.

[2] J. Beau, "Spectral Admissibility under Bounded Flux: Non-Abelian Mode Selection on Cayley Graphs of SU(2) Subgroups," Zenodo (2026). DOI: 10.5281/zenodo.18944985.

[3] J. Beau, "Spectral Capacity Functional and the Binary-Polyhedral Maximality Conjecture," Zenodo (2026). DOI: 10.5281/zenodo.18945273.

[4] J. Beau, "Spectral Gram Rigidity," Zenodo (2026). DOI: 10.5281/zenodo.18969867.

[5] J. Beau, "Three Particle Generations and Mass Hierarchy from Spectral Stratigraphy," Zenodo (2026). DOI: 10.5281/zenodo.19040490.

[6] J. Beau, "Asymptotic Saturation of Projective Resolution: Expander Relaxation Graphs," Zenodo (2026). DOI: 10.5281/zenodo.19056975.

[7] A. Lubotzky, R. Phillips, and P. Sarnak, "Ramanujan graphs," *Combinatorica* **8**, 261–277 (1988).

[8] B. D. McKay, "The expected eigenvalue distribution of a large regular graph," *Linear Algebra Appl.* **40**, 203–216 (1981).

[9] H. Kesten, "Symmetric random walks on groups," *Trans. Amer. Math. Soc.* **92**, 336–354 (1959).